

SABARISH MUTHUMANI

San Jose, CA 95112 (540)605-0766 sabarishm@vt.edu [linkedin.com/in/sabarishmn](https://www.linkedin.com/in/sabarishmn) [sabarishmn.github.io](https://github.com/sabarishmn)

Education

Virginia Polytechnic Institute and State University <i>Master of Science in Computer Engineering</i>	Aug. 2023 – May 2025 <i>Blacksburg, USA</i>
Amrita Vishwa Vidyapeetham <i>Bachelor of Science in Computer Science and Engineering, 7.7/10</i>	Aug. 2016 – May 2020 <i>Coimbatore, India</i>

Technical Skills

Architecture/Interconnects: PCIe Gen4/5, NVMe, DDR4/DDR5, Cache Hierarchy, DMA, x86 & RISC-V Architectures
Hardware Systems: GPU/DGX Platforms, Server Assembly, NIC Firmware, RAID Controllers, SFP+, InfiniBand
Firmware & Bring-Up: BIOS/UEFI Configuration, Power-On Validation, Device Enumeration, BMC/IPMI, GRUB/Kernel Debugging
Validation/Debugging: Root-Cause Analysis, Hardware Diagnostics, Log Analysis, Python/Bash Automation
Programming & Operating Systems: C/C++, Python, Bash, Linux (RHEL/Ubuntu), RTOS

Work Experience

Virginia Tech <i>System Administrator - Network and Server Support</i>	May 2025 - Present (11 months) <i>United States, Remote</i>
--	---

- Created Grafana alerts for PCIe link instability, thermal anomalies, and hardware degradation.
- Restored failed systems by isolating OS corruption, firmware issues, and hardware faults.
- Recovered DGX A100 workstation from crypto-miner compromise by reimaging the OS and performing BIOS/BMC firmware reflash.
- Diagnosed GPU systems with strong understanding of NVLink/NVSwitch interconnect topology in multi-GPU systems.
- Managed out-of-band server operations using BMC(IPMI, Redfish) for remote monitoring, and diagnostics.
- Built Prometheus monitoring for thermals, power, and system health across nodes using dcgm-exporter.
- Validated system bring-up by analyzing BIOS logs, device initialization, and NVMe behavior.
- Reduced downtime by 30% by diagnosing RAID failures and validating disk fault tolerance.
- Improved readiness by 40% by debugging PCIe enumeration, NIC issues, and kernel driver mismatches.
- Worked with GPU memory subsystems (GDDR, HBM) during hardware bring-up and validation.

Virginia Tech <i>Graduate Assistant</i>	August 2024 - May 2025 <i>Blacksburg, Virginia</i>
---	--

- Worked in HPC environments on DGX A100 systems, analyzing PCIe topology and HDR-200 InfiniBand interconnects on AMD EPYC platforms.
- Ran Docker workloads and validated GPU/CPU, memory, and network isolation under high research load.
- Improved system efficiency by tuning memory/cache and benchmarking synchronization on 64-core systems.
- Sped-up deployment by 50% by automating provisioning of 30 RHEL systems with BIOS/firmware tuning.
- Created validation runbooks and RCA docs to standardize onboarding and reduce recurring issues.
- Increased network performance by 20% via VLAN segmentation and TCP/IP validation on Cisco switches.

Cisco Systems, Inc. <i>Software Engineer</i>	April 2022 - August 2023 (1 year 8 months) <i>Bangalore, India</i>
--	--

- Built asynchronous RESTful APIs with multithreading (ThreadPool) and locks for R10, R9, and WebexCC 2.0 customers in Java and Kafka, deployed on Kubernetes, reducing cut-over time by 75%.

National Payments Corporation of India (NPCI) <i>Associate Blockchain</i>	May 2020 - March 2022 (1 year 10 months) <i>Hyderabad, India</i>
---	--

- Automated Kubernetes cluster setup and on-boarding of more than 150 banks to a Hyperledger Fabric FinTech platform using Bash and Python scripts, reducing setup errors by 80%.

PROJECTS

Live Streaming Application For Cricket - [Git]	Dec 2024
---	-----------------

- Built a live streaming app, Winner's Circle Live, using Kafka, AWS RDS and S3 for storage, Apache Airflow for orchestration, while performing Exploratory Data Analysis (EDA) using Pandas and Matplotlib.

Custom Work Stealing Algorithm - [Git]	Mar 2025
---	-----------------

- This project implements and benchmarks two algorithms, the Custom Work Stealing Algorithm and ForkJoinPool in Java.

Interactive RAG Chatbot-GenAI - [Git]	May 2024
--	-----------------

- The primary objective of this project was to develop an interactive chatbot using Azure AI Document Intelligence in real-time using Deepgram, Groq to analyze information from invoice images.